Constant and Free Variables

```
ka = sym('k_a', "rational");
ke = sym('k_e', "rational");
assumeAlso(0 < ka & ka < 1);
assumeAlso(0 < ke & ke < ka);

s = sym('s', "clear");
t = sym('t', "real");
assume(0 <= t);
% time of n-th administration (relative to \delta_n = 0)
tn = sym('t_n', "real");
assume(0 < tn);

dn_eqn(tn) = t - tn;
dn = sym('delta_n', "real");
assumeAlso(dn == dn_eqn(tn));

k = ka / (ke - ka);
kappa = sym('kappa', "rational");
assumeAlso(kappa == k & k < 0 & kappa < 0);

sympref("HeavisideAtOrigin", 1);
sympref("AbbreviateOutput", false);
```

Models

Absorption Model

```
Ca(dn) = symfun('C_a(delta_n)', dn);

% the n-th administration concentration
cn = sym('c_n', "real");
assume(0 < cn);
```

```
ODE_1(dn) = diff(Ca, dn) == -ka * Ca(dn)
```

ODE_1(delta_n) =

$$\frac{\partial}{\partial \delta_n} C_a(\delta_n) = -k_a\, C_a(\delta_n)$$

```
IC_1 = Ca(0) == cn
```

IC_1 = $C_a(0) = c_n$

```
Ca(dn) = dsolve(ODE_1, IC_1)
```

Ca(delta_n) = $c_n\, \mathrm{e}^{-\delta_n k_a}$

Elimination Model

```
C(dn) = symfun('C(delta_n)', dn);
```

```
% initial concentration (in body)
C0 = sym('C_0', "real"); % relative to t = 0
assume(0 <= C0);

C0_n_eqn(tn) = C0 * exp(-ke * tn); % relative to to each \delta_n = 0
C0_n = sym('C_0_n', "real");
assume(0 <= C0_n & C0_n <= C0);
```

```
ODE_2_lhs(dn) = diff(C(dn), dn) - (-ke * C(dn))
```

ODE_2_lhs(delta_n) =

$$\frac{\partial}{\partial \delta_n} C(\delta_n) + k_e\, C(\delta_n)$$

```
IC_2 = C(0) == C0_n
```

IC_2 = $C(0) = C_{0,n}$

Input

```
r_n(dn) = heaviside(dn) * ka * Ca(dn)
```

r_n(delta_n) = $c_n\, k_a\, \mathrm{e}^{-\delta_n k_a}\, \mathrm{heaviside}(\delta_n)$

Solutions

Complementary Solution

```
Cc(dn) = dsolve(ODE_2_lhs(dn) == 0, IC_2)
```

Cc(delta_n) = $C_{0,n}\, \mathrm{e}^{-\delta_n k_e}$

Particular Solution

```
A = sym('A', "real");
Cp(dn) = A*k * heaviside(dn) * ( exp(-ka * dn) - exp(-ke * dn) );
```

```
A = simplify(solve( subs(ODE_2_lhs, C, Cp) == r_n , A ), Steps=50)
```

A = c_n

```
Cp(dn) = simplify(subs(Cp), Steps=50)
```

Cp(delta_n) = $c_n\, \kappa\, \mathrm{heaviside}(\delta_n)\, \left(\mathrm{e}^{-\delta_n k_a} - \mathrm{e}^{-\delta_n k_e}\right)$

General Solution

```
C(dn) = Cc(dn) + Cp(dn)
```

C(delta_n) = $C_{0,n}\,\mathrm{e}^{-\delta_n k_e} + c_n\,\kappa\,\mathrm{heaviside}(\delta_n)\left(\mathrm{e}^{-\delta_n k_a} - \mathrm{e}^{-\delta_n k_e}\right)$

```
C_t(t) = simplify(subs(C, {C0_n, dn}, {C0_n_eqn, dn_eqn}))
```

C_t(t) = $C_0\,\mathrm{e}^{-k_e t} + c_n\,\kappa\,\mathrm{heaviside}(t - t_n)\left(\mathrm{e}^{-k_a (t-t_n)} - \mathrm{e}^{-k_e (t-t_n)}\right)$

Pharmacokinetic Parameters

Area Under the Curve

```
AUC = simplify(int(C_t, [0, Inf]))
```

AUC =

$$\frac{C_0 + c_n}{k_e}$$

Area Under the Moment Curve

```
AUMC = simplify(int(t * C_t, [0 Inf]), Steps=100)
```

AUMC =

$$\frac{C_0 + c_n}{{k_e}^2} + \frac{c_n\,(k_a\,t_n + 1)}{k_a\,k_e}$$

Mean Retention Time

```
MRT = simplify(AUMC / AUC, Steps=100)
```

MRT =

$$\frac{1}{k_e} + \frac{c_n\,(k_a\,t_n + 1)}{k_a\,(C_0 + c_n)}$$

Time for reaching local maximum after n-th administration (relative to \delta_n = 0)

```
cond_max(dn) = diff(subs(C, heaviside(dn), 1), dn) == 0
```

cond_max(delta_n) = $-c_n\,\kappa\left(k_a\,\mathrm{e}^{-\delta_n k_a} - k_e\,\mathrm{e}^{-\delta_n k_e}\right) - C_{0,n}\,k_e\,\mathrm{e}^{-\delta_n k_e} = 0$

```
delta_max_n = (log(ka / ke) - log( 1 - C0_n/(kappa * cn) )) / (ka - ke)
```

delta_max_n =

$$\frac{\log\left(\frac{k_a}{k_e}\right) - \log\left(1 - \frac{C_{0,n}}{c_n\,\kappa}\right)}{k_a - k_e}$$

```
isAlways(simplify(expand( subs(cond_max(delta_max_n), {kappa, C0_n}, {k,
C0_n_eqn(tn)}) ))) % verify
```

```
ans = logical
   1
```

Find k_a from \delta_max_n via Newton's method (when k_e is known)

```
delta_max_n_sym = sym('delta_max_n', "real");
assume(0 < delta_max_n_sym);
```

```
ka_fun(ka) = simplify( (log(ka / ke) - log( 1 - C0_n/(kappa*cn) )) -
delta_max_n_sym * (ka - ke) )
```

ka_fun(k_a) =

$$\log\left(\frac{k_a}{k_e}\right) - \log\left(1 - \frac{C_{0,n}}{c_n\,\kappa}\right) - \delta_{\max,n}\,(k_a - k_e)$$

```
ka_expr(ka) = ka - ka_fun(ka) / diff(ka_fun, ka)
```

ka_expr(k_a) =

$$k_a - \frac{\log\left(1 - \frac{C_{0,n}}{c_n\,\kappa}\right) - \log\left(\frac{k_a}{k_e}\right) + \delta_{\max,n}\,(k_a - k_e)}{\delta_{\max,n} - \frac{1}{k_a}}$$

and also an Empirical Relationship (SSE=0.0028, DFE=1383)

```
x = log(t * ke);
log(ka/ke) == vpa(2.3964) * (exp(vpa(0.4678)*x) - exp(-0.8*x)) ...
    - vpa(0.5248) * ((exp(-x/2) * sin(x) + exp(vpa(0.3842)*x) * x)) ...
    + vpa(0.1330) * exp(-x/1.0559) * sin(exp(x/12.0528) * x) * x;
```

Steady State

- - - New Definitions - - -

```
tau = sym('tau', "real"); % period
assumeAlso(0 < tau & tn < tau);
```

Time when reached steady state (numerical solution, relative to last \delta_n = 0)

```
delta_ss = 5*log(sym(2)) / ke
```

delta_ss =

$$\frac{5\,\log(2)}{k_e}$$

Steady state C_{0} (analytical solution, relative to \delta_n = 0)

```
bc_ss = C0_n == C(tau)
```

bc_ss = $C_{0,n} = C_{0,n}\,\mathrm{e}^{-k_e\,\tau} + c_n\,\kappa\,\left(\mathrm{e}^{-k_a\,\tau} - \mathrm{e}^{-k_e\,\tau}\right)$

```
C_0_ss = simplify( solve(bc_ss, C0_n, IgnoreProperties=true) )
```

C_0_ss =

$$-\frac{c_n\,\kappa\,\left(\mathrm{e}^{-k_a\,\tau} - \mathrm{e}^{-k_e\,\tau}\right)}{\mathrm{e}^{-k_e\,\tau} - 1}$$

Something Fun

Laplace transform of C(t) that repeats with interval \tau

```
L_Css = simplify( ...
    laplace(subs(C_t, heaviside(dn_eqn), 0), s) ...
```

```
    + int( exp(-s*t) * subs(C_t, C0, 0), t, [0 tau] ) / (1 - exp(-s*tau)) ...
    , Steps=100 ...
)
```

L_Css =

$$\frac{C_0}{k_e + s} - \frac{\frac{c_n \kappa \left(e^{-s t_n} - e^{k_a t_n} e^{-k_a \tau} e^{-s \tau}\right)}{k_a + s} - \frac{c_n \kappa \left(e^{-s t_n} - e^{k_e t_n} e^{-k_e \tau} e^{-s \tau}\right)}{k_e + s}}{e^{-s \tau} - 1}$$

Equation for the such system

```
mu_tau(t) = tau * floor(-(t-tau) / tau);
mu_tn(t) = tau * floor(-(t-tn) / tau);
```

```
Css(t) = simplify(piecewise( ...
    0 <= t & t < tn, C_t(t), ...
    tn <= t & t < tau, subs(C_t(t), heaviside(t-tn), 1), ...
    tau <= t, subs(C_t(t), heaviside(t-tn), 0) ...
        + kappa*cn * exp(-ka * (t-tn)) * (exp(-ka * mu_tn(t)) - exp(-ka *
mu_tau(t))) / (exp(ka * tau) - 1) ...
        - kappa*cn * exp(-ke * (t-tn)) * (exp(-ke * mu_tn(t)) - exp(-ke *
mu_tau(t))) / (exp(ke * tau) - 1) ...
))
```

Css(t) =

$$\begin{cases} C_0 e^{-k_e t} & \text{if} \\ C_0 e^{-k_e t} + c_n \kappa \left(e^{-k_a (t-t_n)} - e^{-k_e (t-t_n)}\right) & \text{if} \\ C_0 e^{-k_e t} + \frac{c_n \kappa e^{-k_a (t-t_n)} \left(e^{-k_a \tau \left\lfloor -\frac{t-t_n}{\tau} \right\rfloor} - e^{-k_a \tau \left\lfloor -\frac{t-\tau}{\tau} \right\rfloor}\right)}{e^{k_a \tau} - 1} - \frac{c_n \kappa e^{-k_e (t-t_n)} \left(e^{-k_e \tau \left\lfloor -\frac{t-t_n}{\tau} \right\rfloor} - e^{-k_e \tau \left\lfloor -\frac{t-\tau}{\tau} \right\rfloor}\right)}{e^{k_e \tau} - 1} & \text{if} \end{cases}$$